

To Find the value of r^2 using Machine Learning Regression

1. Multiple Linear Regression (R square value) = 0.8752

2. Support Vector Machine

S.No	Hyper Parameter	Linear(r value)	RBF(Non-Linear r value)	Poly(r value)	Sigmoid (r value)
1.	C10	-0.0396	-0.0566	-0.0536	-0.0547
2.	C100	-0.3205	-0.05741	-0.0574	-0.0572
3.	C1000	0.7802	0.0067	0.2661	0.1850
4.	C2000	0.8767	0.0675	0.4810	0.3970
5.	C3000	0.8956	0.1232	0.6370	0.5913

The SVM Regression use R Square value linear and Hyperparameter(C3000)=0.8956

3. Decision tree

S.No	Creterion	Max Features	Splitter	R value
1.	Friedmanmse	sqrt	random	0.8759
2.	Absolute_error	Auto	random	0.9329
3.	poisson	Auto	random	0.9504
4.	mse	sqrt	best	0.9420
5.	Absolute error	sqrt	best	0.9621
6.	poisson	auto	best	0.9291

The Decision Tree regression use R^2 value (Absolute _ error, Splitter is best=0.9621)

4. Random Forest

S.no	Creterion	Max features	n-Estimators	R_value
1.	Squared_error	None	10	0.9252
2.	Squared_error	None	50	0.9446
3.	Squared_error	None	100	0.9460
4.	Squared_error	None	150	0.9439
5.	Squared_error	None	300	0.9441

Random Forest R^2 value (criterion=squared_error, Max=none, N_estimators=100)=0.9460

The final value of Machine Learning Regression:

Random Forest R^2 value (criterion=squared_error, Max=none, N_estimators=100)=0.9460

The Decision Tree regression use R^2 value (Absolute _ error, Splitter is best=0.9621)

